Communication Systems Lecture-10: Inter-Symbol-Interference

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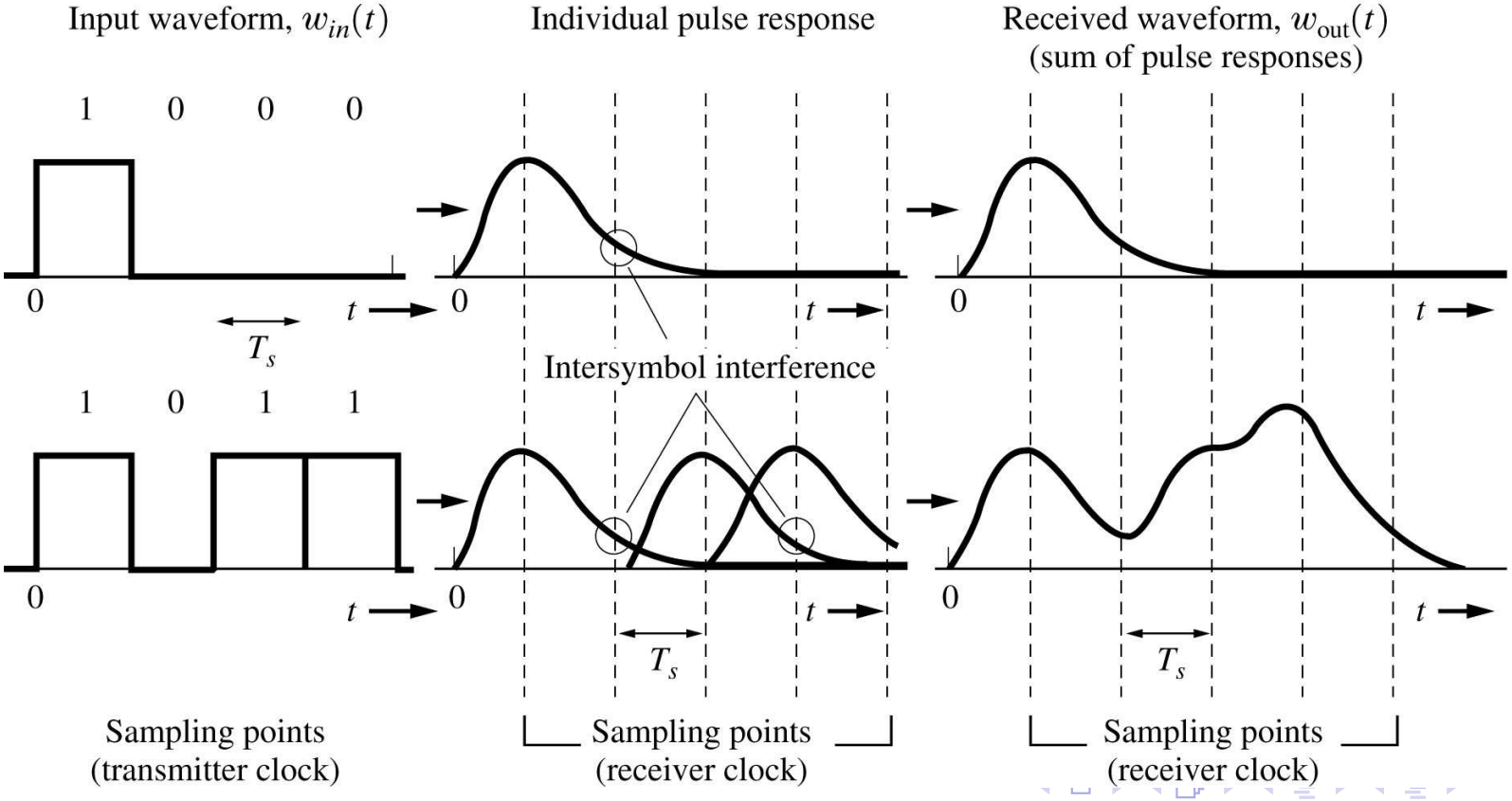
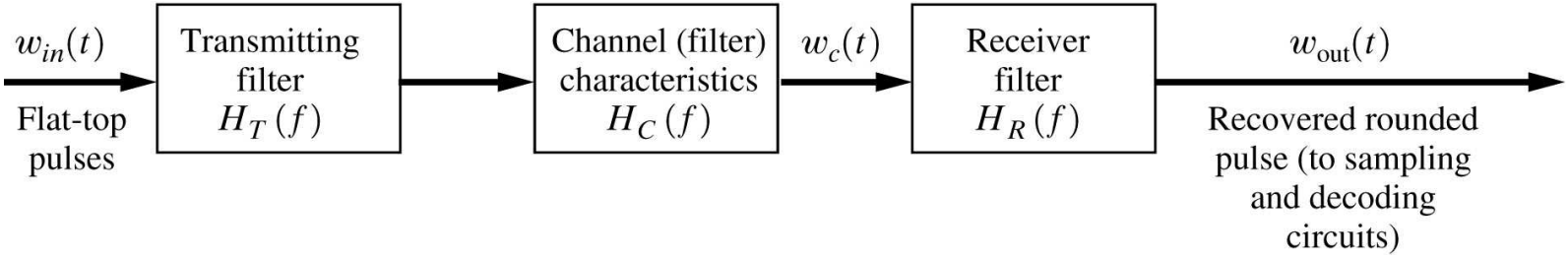
ISI (1)

- The absolute bandwidth of rectangular pulses is infinity.
- When these pulses are transmitted over a communication channel, they get filtered and as a result they spread in time.
- Consequently, pulses corresponding to consecutive symbols will interfere with each other. This is referred to as Inter-Symbol-Interference (ISI).



ISI (2)



- The flat-topped signal at the input can be written as:

$$w_{in}(t) = \sum_n a_n h(t - nT_s)$$

where:
  - $h(t) = \text{rect}\left(\frac{t}{T_s}\right)$.
  - T_s is the symbol duration and the baud rate is $D = \frac{1}{T_s}$.
  - The symbols $\{a_n\}$ can take $L \geq 2$ possible values.
- $w_{in}(t)$ can be written as:

$$w_{in}(t) = \sum_n a_n h(t) * \delta(t - nT_s)$$

$$= \left[\sum_n a_n \delta(t - nT_s)\right] * h(t)$$

ISI (3)

- The output signal can be written as:

$$w_{out}(t) = \left[\sum_n a_n \delta\left(t - nT_s\right)\right] * h_e(t)$$

where $h_e(t)$ is the equivalent impulse response given by:

$$h_e(t) = h(t) * h_T(t) * h_C(t) * h_R(t)$$

where:

  - $h_T(t)$ is the impulse response of the transmitting filter.
  - $h_C(t)$ is the impulse response of the channel.
  - $h_R(t)$ is the impulse response of the receiver filter.

- In the frequency domain:

$$H_e(f) = H(f)H_T(f)H_C(f)H_R(f)$$

where:

$$H(f) = T_s \mathrm{sinc}\left(T_s f\right)$$

ISI (4)

- $h_e(t)$ is the pulse shape that appears at the output of the receiver when the input is a flat-top pulse.
- In other words, the signal at the output of the receiver can be written as:

$$w_{out}(t) = \sum_n a_n h_e(t - nT_s)$$

- Since we can not control the impulse response of the channel, the problem is to determine the transmitting filter $H_T(f)$ and the receiver filter $H_R(f)$ that will minimize the impact of ISI.
- ISI can be eliminated by applying Nyquist's first method (zero ISI).

Nyquist's first method (1)

- Nyquist's first method for eliminating ISI consists of choosing an equivalent impulse response $h_e(t)$ that verifies the following relation:

$$h_e(kT_s) = \begin{cases} C, & k = 0; \\ 0, & k \neq 0. \end{cases}$$

  where k is an integer and C is a constant.
- Since: $w_{out}(t) = \sum_n a_n h_e(t - nT_s)$, at the time instant jT_s (j-th symbol duration):

$$\begin{aligned} w_{out}(jT_s) &= \sum_n a_n h_e((j-n)T_s) \\ &= a_j h_e(0) + \underbrace{\sum_{n \neq j} a_n h_e((j-n)T_s)}_{ISI} = Ca_j \end{aligned}$$

  where the last equation follows when Nyquist's first method is satisfied. In this case, the j-th sample of the output signal is proportional to the j-th symbol a_j.

Nyquist's first method (2)

- $h_e(t)$ can be chosen to take the following value:

$$h_e(t) = \mathrm{sinc}\,(f_s t)$$

- This filter satisfies Nyquist's first method since:

$$h_e(kT_s) = \mathrm{sinc}\,(k) = \begin{cases} 1, & k = 0; \\ 0, & k \neq 0. \end{cases}$$

- However, this choice of $h_e(t)$ has 2 practical difficulties:
  - $h_e(t)$ is non-causal ($h_e(t) \neq 0$ for $t < 0$).
  - The tail of $\mathrm{sinc}(t)$ decays as $\frac{1}{t}$. Consequently, inaccurate synchronization will result in excessive ISI.
- One solution for the choice of the equivalent filter is the raised cosine-rolloff Nyquist filter.

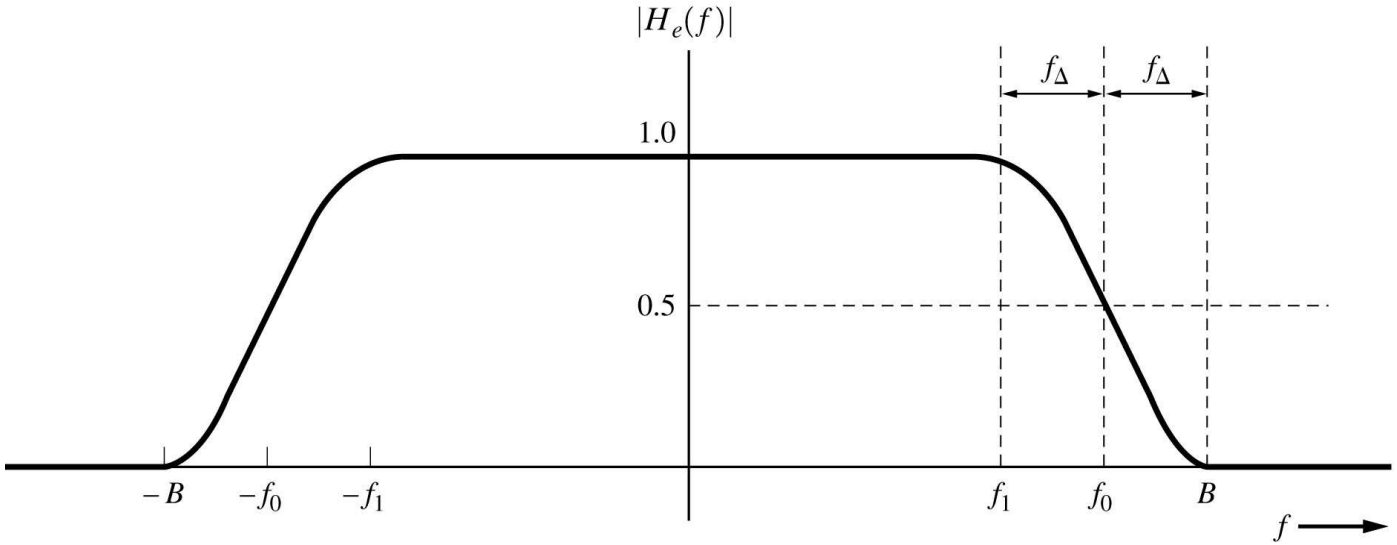
Raised Cosine-Rolloff Nyquist Filter (1)

- The raised cosine-rolloff Nyquist filter has the following transfer function:

$$H_e(f) = \begin{cases} 1, & |f| < f_1; \\ \frac{1}{2}\left\{1 + \cos\left[\frac{\pi(|f|-f_1)}{2f_\Delta}\right]\right\}, & f_1 < |f| < B; \\ 0, & |f| > B. \end{cases}$$

where B is the absolute bandwidth and:

$$f_\Delta = B - f_0 = f_0 - f_1$$



Raised Cosine-Rolloff Nyquist Filter (2)

- The rolloff factor is defined as:

$$r \triangleq \frac{f_\Delta}{f_0}$$

- The 6-dB bandwidth of the filter is equal to f_0:

$$\begin{aligned} H_e(f_0) &= \frac{1}{2}\left\{1+\cos\left[\frac{\pi\left(f_0-f_1\right)}{2f_\Delta}\right]\right\} \\ &= \frac{1}{2}\left\{1+\cos\left[\frac{\pi f_\Delta}{2f_\Delta}\right]\right\} = \frac{1}{2}\left\{1+\cos\left[\frac{\pi}{2}\right]\right\} = \frac{1}{2} \end{aligned}$$

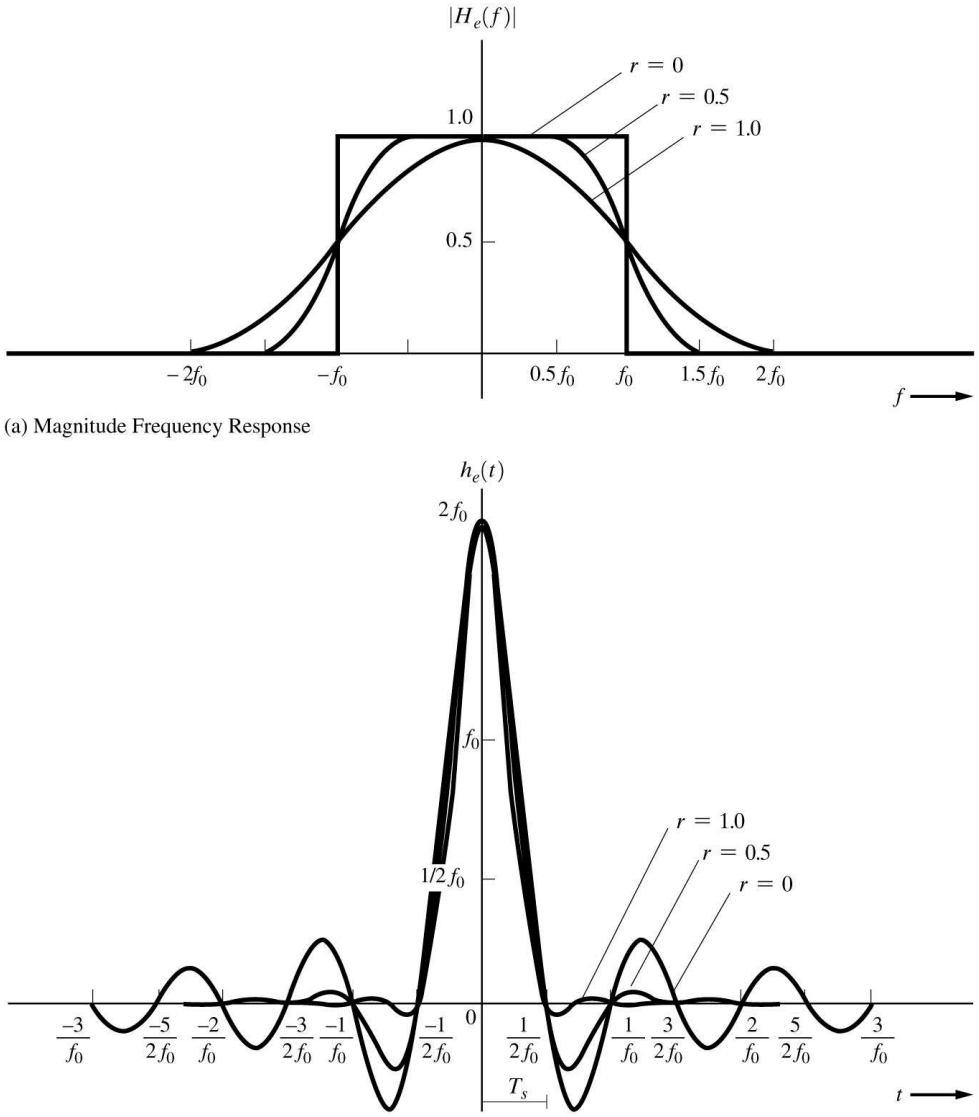
- The corresponding impulse response is:

$$h_e(t) = 2f_0\mathrm{sinc}\left(2f_0 t\right)\left[\frac{\cos(2\pi f_\Delta t)}{1-(4f_\Delta t)^2}\right]$$

- Note that:
  - $h_e(t) = 0$ at nonzero integer multiples of $\frac{1}{2f_0}$.
  - For $r = 0$, $f_\Delta = 0$ and $h_e(t) = 2f_0\mathrm{sinc}\left(2f_0 t\right)$.

Raised Cosine-Rolloff Nyquist Filter (3)

- Inaccurate synchronization will cause less ISI since, for $r \neq 0$, $h_e(t)$ decays as $\frac{1}{t^3}$ rather than $\frac{1}{t}$.



(a) Magnitude Frequency Response

Raised Cosine-Rolloff Nyquist Filter (4)

- Since $h_e(t) = 0$ at nonzero integer multiples of $\frac{1}{2f_0}$, then Nyquist's first method for eliminating ISI is satisfied when the symbol duration verifies $T_s = \frac{1}{2f_0}$.
- The corresponding baud rate is:

$$D \triangleq \frac{1}{T_s} = 2f_0$$

  the baud rate is twice the 6-dB bandwidth.
- On the other hand:

$$r \triangleq \frac{f_\Delta}{f_0} = \frac{B - f_0}{f_0} \quad \Rightarrow \quad B = (1 + r)f_0$$

- Therefore, for a given absolute bandwidth B and rolloff factor r, the baud rate that can be supported by the system is:

$$D = \frac{2B}{1 + r}$$

- The raised-cosine filter belongs to a class of filters that satisfy Nyquist's first criterion. These are called Nyquist filters.

Raised Cosine-Rolloff Nyquist Filter (5)

- Note that raised cosine-rolloff filters are not causal.
- To solve this problem a delay of $T_d > 0$ can be introduced. In this case, $h_e(t - T_d)$ will be approximately causal if T_d is sufficiently larger than $\frac{1}{2f_0}$.
- This additional delay will only introduce a linear phase shift since:

$$\mathcal{F}\left[h_e(t - T_d)\right] = H_e(f)e^{-j2\pi T_d f}$$

- In other words, the magnitude of the equivalent transfer function will remain the same.

Nyquist Filters (1)

THEOREM: A filter is said to be a Nyquist filter if the effective transfer function is:

$$H_e(f) = \begin{cases} \mathrm{rect}\left(\frac{f}{2f_0}\right) + Y(f), & |f| < 2f_0; \\ 0, & \text{elsewhere.} \end{cases}$$

where $Y(f)$ is a real function:

- It is even symmetric about $f = 0$:

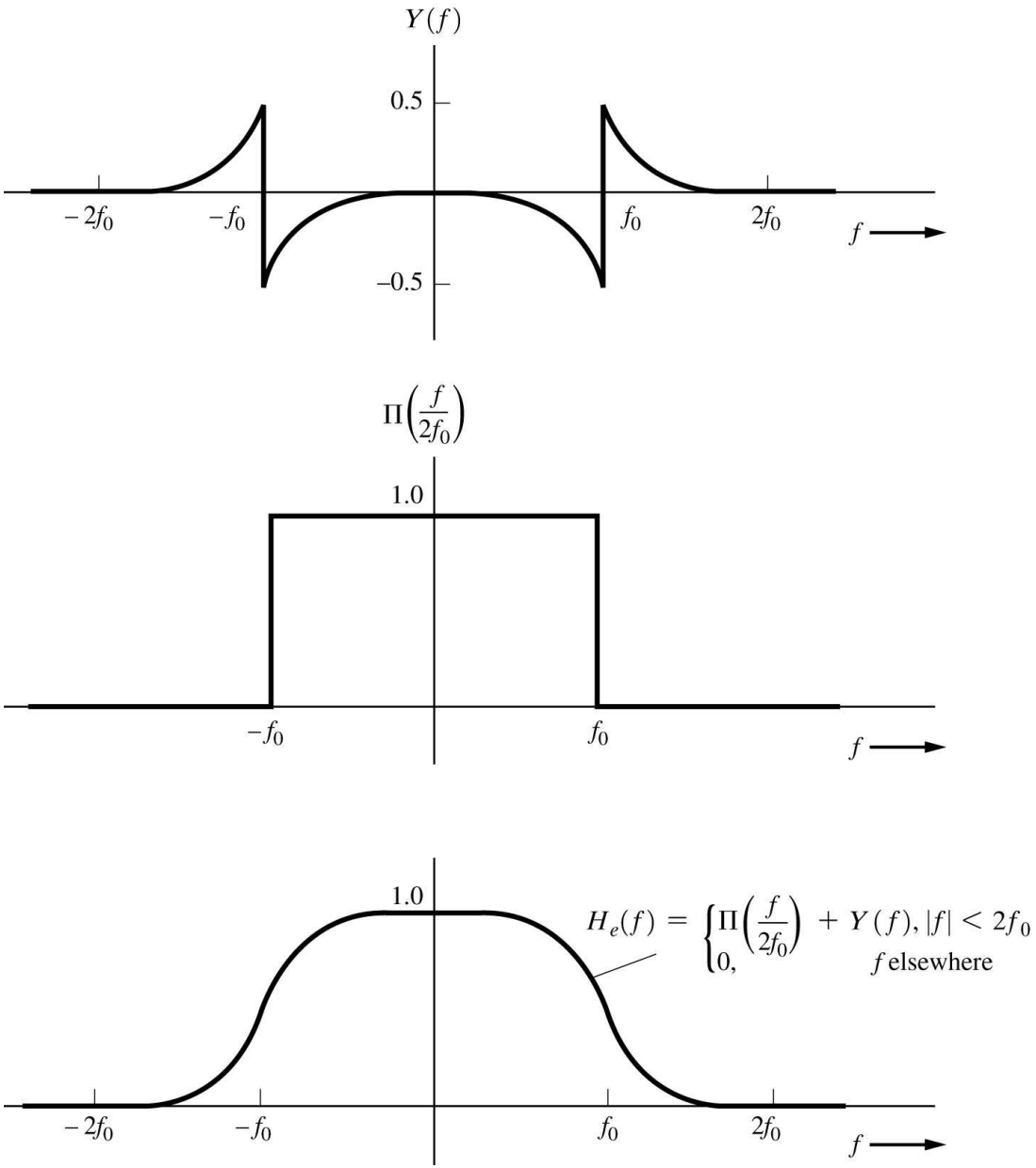
$$Y(-f) = Y(f) \quad ; \quad |f| < 2f_0$$

- It is odd symmetric about $f = f_0$:

$$Y(-f + f_0) = -Y(f + f_0) \quad ; \quad |f| < f_0$$

There will be no ISI when the symbol duration verifies: $T_s = \frac{1}{2f_0}$.

Nyquist Filters (2)



Nyquist Filters (3)

- In order to prove the last theorem, we must prove that $h_e(nT_s) = 0$ when n is a nonzero integer.
- Taking the inverse Fourier transform of $H_e(f)$:

$$h_e(t) = \int_{-\infty}^{+\infty} H_e(f) e^{j2\pi ft} \mathrm{d}f = \int_{-\infty}^{+\infty} \left[\mathrm{rect}\left(\frac{f}{2f_0}\right) + Y(f) \right] e^{j2\pi ft} \mathrm{d}f$$

$$= \int_{-f_0}^{+f_0} e^{j2\pi ft} \mathrm{d}f + \int_{-2f_0}^{+2f_0} Y(f) e^{j2\pi ft} \mathrm{d}f$$

$$= 2f_0 \mathrm{sinc}\,(2f_0 t) + \int_{-2f_0}^{0} Y(f) e^{j2\pi ft} \mathrm{d}f + \int_{0}^{+2f_0} Y(f) e^{j2\pi ft} \mathrm{d}f$$

- Let $f_1 = f + f_0$ in the first integral and $f_1 = f - f_0$ in the second integral:

$$h_e(t) = 2f_0 \mathrm{sinc}\,(2f_0 t) + e^{-j2\pi f_0 t} \int_{-f_0}^{f_0} Y(f_1 - f_0) e^{j2\pi f_1 t} \mathrm{d}f_1$$

$$+ e^{j2\pi f_0 t} \int_{-f_0}^{f_0} Y(f_1 + f_0) e^{j2\pi f_1 t} \mathrm{d}f_1$$

Nyquist Filters (4)

- Given that $Y(f_1 - f_0) = Y(-f_1 + f_0) = -Y(f_1 + f_0)$:

$$h_e(t) = 2f_0 \mathrm{sinc}\,(2f_0 t) + \left[e^{j2\pi f_0 t} - e^{-j2\pi f_0 t}\right] \int_{-f_0}^{f_0} Y(f_1 + f_0) e^{j2\pi f_1 t} \mathrm{d}f_1$$

$$= 2f_0 \mathrm{sinc}\,(2f_0 t) + 2j \sin(2\pi f_0 t) \int_{-f_0}^{f_0} Y(f_1 + f_0) e^{j2\pi f_1 t} \mathrm{d}f_1$$

- For $t = nT_s = \frac{n}{2f_0}$:

$$h_e(nT_s) = 2f_0 \mathrm{sinc}\,(n) + 2j \sin(\pi n) \int_{-f_0}^{f_0} Y(f_1 + f_0) e^{j2\pi f_1 t} \mathrm{d}f_1$$

$$= 2f_0 \mathrm{sinc}\,(n) = \begin{cases} 2f_0, & n = 0; \\ 0, & n \neq 0. \end{cases}$$

- Consequently, $h_e(t)$ satisfies Nyquist's first criterion.
- Note that Nyquist filters verify the relation $H_e(-f) = H_e^*(f)$ resulting in a real impulse response $h_e(t)$.

Nyquist Filters (5)

- Given an equivalent transfer function $H_e(f)$ that results in an impulse response that satisfies Nyquist's first criterion, the filters at the transmitter and the receiver can be chosen as:

$$|H_T(f)| = \frac{\sqrt{|H_e(f)|}}{|H(f)|\sqrt{|H_C(f)|}}$$

$$|H_R(f)| = \frac{\sqrt{|H_e(f)|}}{\sqrt{|H_C(f)|}}$$

- If the channel has a flat frequency response and $H(f)$ is flat over the bandwidth of interest ($h(t)$ is a sinc-type function):

$$|H_T(f)| = |H_R(f)| = \sqrt{|H_e(f)|}$$

In this case, if $H_e(f)$ is a raised cosine-rolloff filter, the filters at the transmitter and the receiver are called square-root raised cosine-rolloff filters.